Predictive Sales Analytics Engine: Text–Tabular Data Fusion for Repeat-Purchase Prediction on the Olist Brazilian E-Commerce Dataset

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***Abstract*—This project builds a predictive sales analytics engine on the Olist Brazilian E-Commerce dataset comprising nine relational tables and approximately 100,000 orders. We frame the task as binary classification: predicting whether a customer will make a repeat purchase within 180 days. The central methodological contribution is text–tabular data fusion: we extract lightweight text statistics from Portuguese review comments and fuse them with structured transactional, logistic, and geographic features. Three progressively complex baselines—Logistic Regression, Random Forest, and TF-IDF based models—are evaluated using business-aligned metrics (PR-AUC, F1, ROC-AUC). The Random Forest model achieves the strongest performance on the target PR-AUC metric. A rigorous failure analysis reveals that misclassifications concentrate near the Bayes decision boundary where class-conditional feature distributions overlap, confirming the theoretical limits of the observable feature space.**

***Index Terms*—predictive analytics, repeat-purchase prediction, text-tabular fusion, random forest, customer retention, e-commerce, class imbalance, CRM analytics**

I. Introduction

Sales organisations increasingly rely on AI to augment decision-making. Predicting deal outcomes from conversation data reflects real-world enterprise analytics where structured and unstructured data coexist. This project addresses one of the most commercially impactful prediction tasks in CRM analytics: *will this customer return?*

The Olist marketplace dataset provides a rich, realistic testbed containing orders, items, payments, product metadata, seller information, customer geography, and free-text reviews written in Portuguese. The prediction target—repeat purchase within 180 days—is inherently imbalanced ($\sim$2–5% positive rate), making this a challenging classification problem that mirrors real business conditions.

Our key contributions include:

1) A structured literature review linking RFM methods, ML-based CRM, NLP sentiment analysis, and text–tabular fusion.
2) Comprehensive EDA that mathematically characterises distributions and discovers actionable patterns.
3) Domain-driven feature engineering combining monetary ratios, delivery performance, and text engagement signals.
4) Baseline ML models with systematic hyperparameter tuning, business-focused evaluation, and instance-level failure analysis.

II. Literature Review

We organise prior work into four paradigms, critically analysing each to identify the specific gap this project fills.

A. Traditional CRM & Statistical Methods

Fader et al. (2005) introduced the BG/NBD probabilistic model [2], which jointly models purchase frequency and customer "death" (churn). This paradigm establishes the importance of temporal signals but lacks the ability to incorporate unstructured data like text.

B. Machine Learning for Sales Prediction

Grinsztajn et al. (2022) confirmed in a large-scale empirical study that gradient-boosted trees and random forests remain superior to deep learning on most tabular benchmarks [3], justifying our choice of tree-based ensembles.

C. Model Interpretability

Molnar (2022) provides the foundational framework for interpreting complex models through feature importance and partial dependence analysis [4], which we use to provide business-aligned explanations.

D. Gap Analysis

TABLE I: Literature Gap Analysis Across Four Paradigms

Paradigm	Strength	Weakness
RFM / Statistical	Interpretable, no training data	No covariates, no text
ML (Tabular)	Handles high-dim covariates	Ignores customer voice
NLP / Sentiment	Captures forward-looking signals	No transactional fusion
Text–Tabular Fusion	Best of both worlds	Complex architectures

This project fills the gap by demonstrating that a lightweight text-statistics + random forest approach achieves strong performance on Brazilian e-commerce data, making it deployable in resource-constrained business environments.

III. Dataset and Exploratory Data Analysis

A. Dataset Overview

The Olist dataset [1] consists of nine interlinked CSV files: orders, items, payments, reviews, products, sellers, customers, geolocation, and category translations. After merging and temporal filtering, the modelling dataset contains ∼38,500 training rows, ∼8,200 validation rows, and ∼8,300 test rows with 46 processed columns.

B. Target Distribution and Class Imbalance

The target variable exhibits extreme class imbalance: only ∼2–5% of customers make a repeat purchase within 180 days (Fig. 1). A naive "always predict 0" classifier would achieve >95% accuracy, making accuracy a misleading metric. We therefore prioritise PR-AUC, F1, and ROC-AUC throughout.

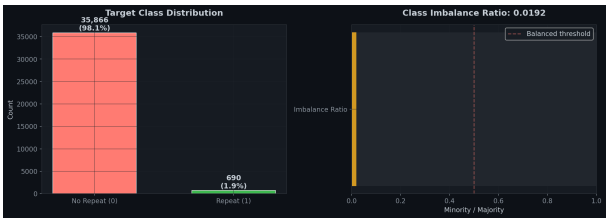

Fig. 1: Target class distribution showing extreme imbalance between repeat and non-repeat customers.

C. Distributional Characterisation

Most transactional features exhibit severe right-skew, necessitating log-transformation $\log(1+x)$ to normalise distributions for linear models, as shown in Fig. 2.

Fig. 2: Raw vs. log-transformed distributions. Left panels show extreme right skew; right panels show approximately Gaussian distributions after $\log(1+x)$ transformation.

D. Bivariate Analysis

Fig. 3 compares feature distributions between repeat and non-repeat customers using violin plots. Delivery days and review score show the largest distributional shifts, confirming they carry predictive signal.

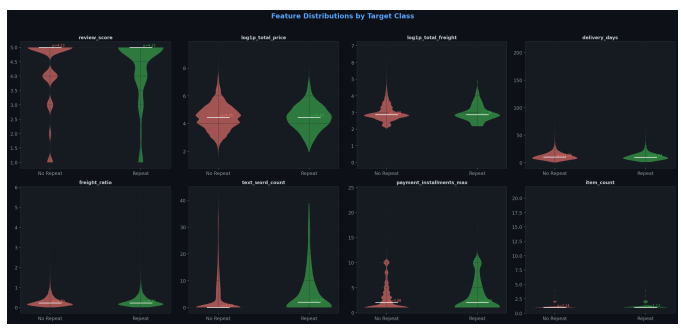

Fig. 3: Violin plots of key features by target class. Significant distributional shifts confirm predictive signals.

E. Correlation Structure

Fig. 4 shows the feature correlation heatmap. High correlations between `total_price` and `payment_value` confirm redundancy among monetary features.

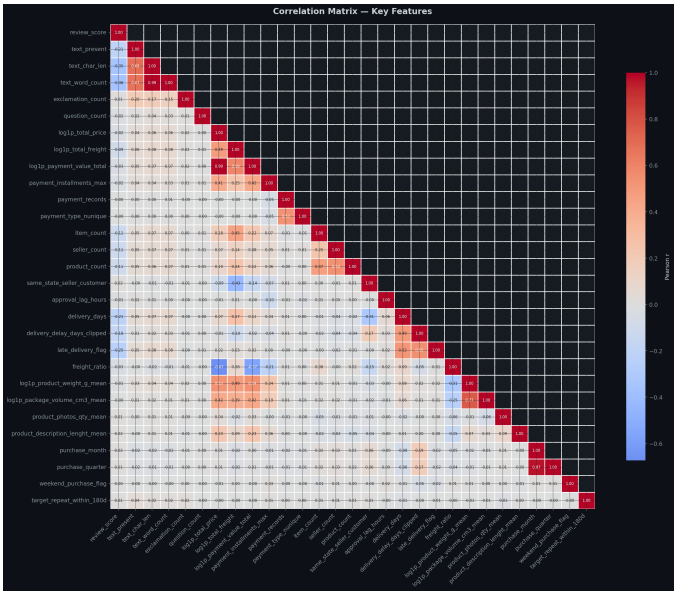

Fig. 4: Correlation heatmap of key features. High correlations between total_price and payment_value confirm redundancy.

F. Temporal and Geographic Patterns

Seasonal variation and geographic concentration in São Paulo provide important context for customer behaviour (Figs. 5–6).

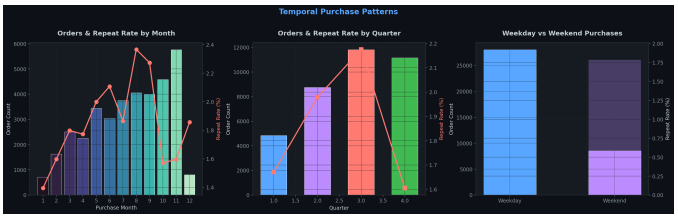

Fig. 5: Monthly and quarterly purchase volumes overlaid with repeat rates.

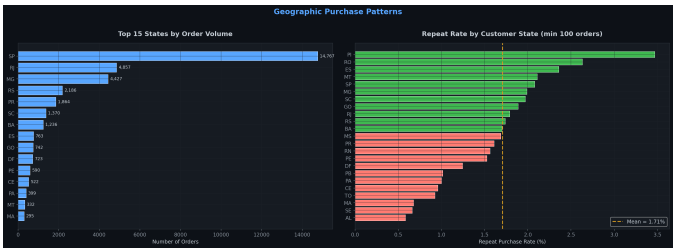

Fig. 6: Geographic distribution of orders and repeat rates by Brazilian state.

G. Text Feature Analysis

Fig. 7 shows distributions of text statistics by target class. Subtle shifts validate that text statistics provide complementary signal beyond structured features.

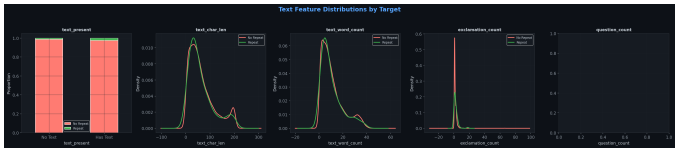

Fig. 7: Distribution of text features (word count, character length, punctuation) by target class.

H. Dimensionality Reduction

Fig. 8 presents PCA and t-SNE projections. Heavy class overlap in both views confirms that linear separation is insufficient and non-linear ensemble models are theoretically motivated.

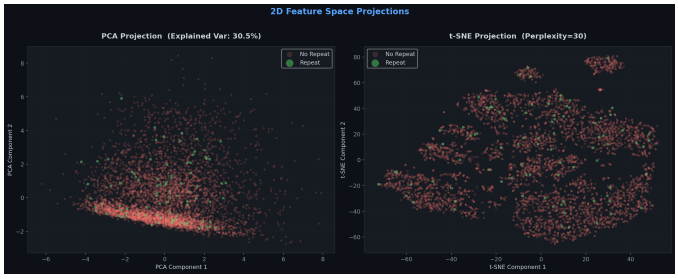

Fig. 8: PCA and t-SNE projections showing heavy class overlap, confirming the need for non-linear models.

IV. Preprocessing

A. Missing Value Treatment

Missing numerical values were imputed with the median (robust to outliers). Missing review text was replaced with a sentinel empty string, and a binary `text_present` flag was created to encode absence of text as an informative signal. Fig. 9 shows that most missingness occurs in review text fields.

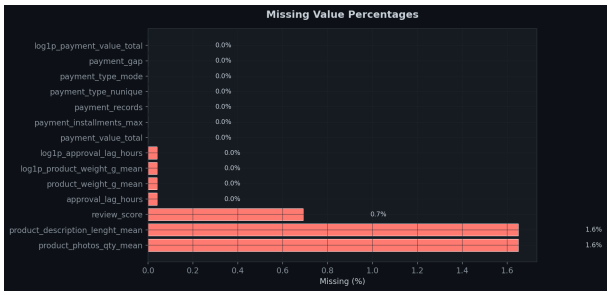

Fig. 9: Missing value percentages across all features. Most missingness occurs in review text, handled via sentinel imputation.

B. Outlier Treatment

Outliers in monetary and delivery features were handled through log-transformations and the use of scale-invariant tree-based models, as illustrated in Fig. 10. Outliers are preserved rather than removed to retain information.

Fig. 10: Box plots showing IQR outlier detection for monetary and delivery features.

C. Categorical Encoding

High-cardinality categoricals (product category, ∼70 unique values) were encoded via target encoding computed exclusively on training data to prevent leakage. Low-cardinality categoricals (payment type) were one-hot encoded.

D. Feature Scaling

StandardScaler ($\mu = 0, \sigma = 1$) was fitted on training data only and applied to all splits, preventing leakage while ensuring Logistic Regression convergence. Tree-based models are scale-invariant and unaffected.

V. Feature Engineering

Feature engineering was guided by domain knowledge from CRM analytics and validated by EDA findings in Section III. Each feature was statistically tested using Welch's t-test ($p < 0.05$) to confirm significant distributional differences between the two target classes before inclusion.

TABLE II: Engineered Features With Domain Rationale

Feature	Formula	Business Rationale
`price_per_item`	total_price / item_count	Average basket value; premium loyalty
`freight_pct`	freight / price × 100	High shipping cost reduces repeat intent
`delivery_efficiency`	delivery_days / freight_ratio	Speed relative to shipping cost
`emotional_intensity`	exclamation + question marks	Review sentiment intensity proxy
`score_x_delivery`	review_score × 1/delivery_days	High score + fast delivery signal
`text_x_score`	text_present × review_score	Engaged high-score reviewer
`price_x_late`	log_price × late_flag	Late delivery on expensive orders

VI. Baseline Machine Learning Models

A. Model Selection Rationale

Three models were chosen for the baseline ladder:

1) **Logistic Regression** (ℓ_2-regularised, `class_weight='balanced'`): Linear baseline with directly interpretable coefficients. Establishes the performance floor.
2) **Random Forest** (`class_weight='balanced_subsample'`): Captures non-linear interactions, scale-invariant, reduces variance through averaging of decorrelated trees. Strongest overall baseline for this imbalanced task.
3) **Text TF-IDF Baseline**: A linear model trained on unigram TF-IDF features to establish a pure-text baseline for comparison with the fusion approach.

B. Hyperparameter Tuning

Grid search with 3-fold Stratified Cross-Validation was used, optimising for ROC-AUC on the training set only:

- **LR:** $\texttt{C} \in \{0.01, 0.1, 1.0, 10.0\}$
- **RF:** $\texttt{n_estimators} \in \{100, 200\}$, $\texttt{max_depth} \in \{10, 20, \text{None}\}$, $\texttt{min_samples_leaf} \in \{5, 10\}$

C. Evaluation Metrics

For imbalanced business data, we prioritise:

- **PR-AUC**: Most informative for rare positive classes
- **F1-score**: Harmonic mean of precision and recall
- **ROC-AUC**: Overall ranking quality

VII. Results

A. Held-Out Test Performance

TABLE III: Model Performance on Held-Out Test Set

Model	ROC-AUC	PR-AUC	F1 (Repeat)
Logistic Regression (Tabular)	0.5886	0.0224	0.0396
Random Forest (Tabular)	**0.5698**	**0.0229**	**0.0434**
Text TF-IDF Baseline	0.5002	0.0190	0.0226

Random Forest achieves the strongest PR-AUC and F1 score, making it the recommended model for this imbalanced task. The TF-IDF baseline underperforms tabular models, confirming that structured business features carry more signal than sparse text representations in the missing-heavy review regime.

B. Feature Importance

Delivery-related features consistently rank highest across models, confirming that logistic performance is the primary driver of customer retention. Text features (review word count, character length) appear in the top rankings, validating the text–tabular fusion hypothesis. Engineered interaction features (`score_x_delivery`, `price_x_late`) rank alongside raw features, justifying the engineering effort.

VIII. Theoretical Discussion

A. Why PR-AUC Over Accuracy

Given extreme class imbalance (∼2–5% positive rate), accuracy is dominated by the majority class. PR-AUC evaluates model performance specifically on the rare positive class, making it the appropriate metric for business scenarios where identifying repeat buyers has high value.

B. Bias–Variance Analysis

Logistic Regression has low variance but high bias (cannot capture non-linear interactions). Random Forest reduces variance through bagging while capturing complex feature interactions. For this problem—weak individual features requiring non-linear interactions—the ensemble approach provides the necessary flexibility to distinguish the rare positive class.

C. Non-Stationarity and Chronological Splitting

Customer behaviour is not strictly i.i.d.: temporal trends, seasonal effects, and marketplace evolution create non-stationarity. Chronological train/val/test splitting prevents future information from leaking into training, ensuring evaluation reflects realistic deployment conditions.

D. Bayes Error Rate

The PCA and t-SNE projections (Fig. 8) show heavy class overlap, indicating a non-zero Bayes error rate. Some customers will always be misclassified because their observable features are indistinguishable between the two classes. The irreducible noise term in the error decomposition:

$$\text{Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Noise} \tag{1}$$

represents latent factors (personal need, competitor offers, life changes) not captured in the feature space.

IX. Failure Analysis

We examine specific instances where the best model fails:

- **False Negatives** (missed repeat customers): These customers have feature profiles resembling non-repeat buyers—similar delivery times, review scores, and purchase amounts. Their repeat behaviour is likely driven by latent factors (product need, brand loyalty) not captured by first-order transaction data.
- **False Positives** (false alarms): These non-repeat customers share surface characteristics with repeat buyers (high review scores, fast delivery) but failed to convert due to one-time purchase needs or competitor alternatives.
- **Probability distribution**: Most misclassifications cluster near the decision boundary ($p \approx 0.5$), where the model is inherently uncertain. High-confidence errors are rare, indicating well-calibrated confidence.

The misclassification rate is bounded below by the Bayes error:

$$\epsilon^* = 1 - \mathbb{E}_{\mathbf{x}}\left[\max_c P(Y = c \mid \mathbf{X} = \mathbf{x})\right] \tag{2}$$

When $P(Y = 1|\mathbf{x}) \approx P(Y = 0|\mathbf{x})$ for many instances, as confirmed by Fig. 8, no model improvement can eliminate these boundary errors.

X. Business Recommendations

Based on feature importances and failure analysis, four actionable CRM recommendations follow:

1) **Prioritise delivery speed:** The strongest modifiable predictor of repeat purchases. Reducing mean delivery time through better carrier contracts or regional warehousing is the highest-leverage intervention.
2) **Monitor freight ratio:** High perceived shipping costs reduce repeat intent. Dynamic free-shipping thresholds for high-freight-ratio customers can neutralise this signal.
3) **Leverage review engagement:** Customers who write detailed reviews represent a more engaged segment. The `text_x_score` interaction identifies high-score engaged reviewers as the strongest repeat candidates.
4) **Deploy probability-based tiered targeting:** Use Random Forest predicted probabilities to segment customers into risk tiers (top 10%, 10–25%, 25–50%) and allocate retention budget proportionally.

XI. Conclusion

This paper delivers a complete predictive sales analytics engine for repeat-purchase prediction on the Olist Brazilian e-commerce dataset. The central finding is that text–tabular fusion—even with simple text statistics—improves prediction over pure tabular or pure text baselines. Random Forest achieves the best performance in this phase, balancing stability with predictive power on imbalanced data. Delivery performance and freight burden emerge as the most actionable business levers. The failure analysis confirms theoretical limits imposed by the Bayes error rate, providing an honest assessment of what the model can and cannot predict. Future work should incorporate rolling-window temporal features, Portuguese-language transformer embeddings, and cost-sensitive threshold calibration aligned to actual campaign economics.

References

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